

AI Assistance Acknowledgement

Portions of the codebase in this project were developed with the assistance of ChatGPT as a programming and learning support tool.

Specifically, ChatGPT was used to:

Assist in optimizing and structuring implementations of the MLP and Word2Vec (from-scratch) models, where the large scale of the AG News dataset required guidance on efficient training pipelines and practical implementation details.

Provide implementation guidance for the BERT Transformer training pipeline, including data preprocessing, tokenization, and model wiring, as Transformer-based architectures were new to me at the time of development.

It is important to note that:

The theoretical understanding and conceptual foundations of NLP models—particularly Bag-of-Words, TF-IDF, Word2Vec, neural language models, and Transformers—were developed through independent study of **Speech and Language Processing** by Dan Jurafsky and James H. Martin, along with course lectures of **Stanford CS224N Natural Language Processing** and standard ML references.

ChatGPT was used as a supplementary aid, similar to documentation or coding forums, to translate understood concepts into correct and efficient code, not as a substitute for conceptual learning or experimental design.

All experiments, architectural decisions, evaluations, and result interpretations were designed, executed, and validated by me.

Use of Grammarly:

Grammarly was used as an extension while writing the Project Report and Readme.md files. Grammarly helped me follow the writing flow of the paragraph and to ensure effective and efficient form of writing and communication.

**** This disclosure is made in the interest of transparency and responsible use of AI-assisted development tools in accordance with academic integrity guidelines.**